

7. _____ are the arguments passed to a function in correct positional order.
A) Required arguments B) Keyword arguments
C) Default arguments D) Variable-length arguments
8. What will be the output of the following code snippet?
`def add(a, b=10):
 return a + b
result = add(5)
print(result)`
A) 5 B) 10
C) 15 D) Error
9. What will be the output of the following Python code?
`str1="helloworld"
print(str1[: -1])`
A) dlrowolleh B) hello
C) world D) helloworld
10. Which of the following methods is used to add an element to the end of a list in Python?
A) append() B) add()
C) insert() D) extend()
11. What is the output of `my_tuple = (1, 2, 3) * 2`?
A) (1, 2, 3, 1, 2, 3) B) (2,4,6)
C) [1,2,3,1,2,3] D) Error
12. What will be the output of `len({'apple': 1, 'banana': 2, 'cherry': 3})`?
A) 1 B) 2
C) 3 D) Error
13. Which of the following file extensions represents a text file?
A) .csv B) .png
C) .txt D) .mp4
14. What is the primary use of the append ('a') mode in file handling?
A) To delete the file content B) To write new content to the file and erase old content
C) To add new content to the end of the file D) To read the file content
15. What is the default mode in which a file is opened using the `open()` function in Python?
A) Write mode ('w') B) Read mode ('r')
C) Append mode ('a') D) Binary mode ('rb')
16. What is the purpose of the `csv.reader()` method in Python?
A) To write data to a CSV file B) To read data from a CSV file
C) To delete data from a CSV file D) To rename a CSV file
17. Which block is executed if no exception occurs within the try block?
A) except B) finally
C) else D) raise
18. What will the output of the following code be?
`try:
 x = 1/0
except ZeroDivisionError:
 x = 0
 print(x)`
A) 1 B) 0
C) An exception will be raised D) 0.5
19. Which Python library is used for data visualization?
A) NumPy B) Pandas
C) Matplotlib D) SciPy
20. What type of plot is created using `plt.plot(x, y)`?
A) Bar chart B) Line plot
C) Scatter plot D) Histogram

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain while loop with syntax and programming example.	4	L*2	1	1
b)	What is module? Explain the various methods of importing modules in Python programs.	8	L2	2	1
c)	Write a python program to find the sum of digits and number of occurrences of digit in a number.	4	L3	1	2
2. a)	What is a variable? What are the rules for writing an identifier?	4	L1	1	1
b)	List the different types of operators. Explain the following operators with an example program.				
i.	Logical Operators	8	L2	1	1
ii.	Identity Operators				
c)	Define recursive functions. Develop a python program to find factorial of a number using recursion.	4	L3	2	2
3. a)	Explain nested if conditional statement in python with syntax and programming example.	4	L2	1	1
b)	Explain different type of arguments used in python functions with example program.	8	L2	2	1
c)	Explain membership operators with a programming example.	4	L2	1	1
Unit – II					
4. a)	Define file. Illustrate read and write operation in file with a code snippet.	4	L2	4	1
b)	Explain the following methods in list with an example: i) len() ii) insert() iii) pop() iv) sort()	8	L2	3	1
c)	Write a Python program that takes a sentence as input from the user, calculates and prints the total number of characters in the sentence, and creates a dictionary containing the frequency of letters and digits in the sentence.	4	L3	3	2
5. a)	Write a Python program to read a list of n integers (positive as well as negative). Create 2 new lists, one having all positive numbers and others having all negative numbers from the given list. Print all 3 lists.	4	L3	3	2
b)	Explain the following methods in a dictionary with example: i) keys() ii) values() iii) items() iv) copy()	8	L2	3	1
c)	Write a program to create a CSV file students.csv and store the rows: ["Name", "Age", "Grade"], ["Alice", 23, "A"], ["Bob", 21, "B"] using Python's csv module.	4	L3	4	2
6. a)	How is a tuple different from a list and what function is used to convert list to tuple? Explain.	4	L2	3	1
b)	Write a python program to perform binary search on a list of integers.	8	L3	4	2
c)	Write the output of the following Python code: spam= 'Hello, World!' i) spam[1:6:2] ii) spam[4:-12:-1] iii) spam[5:0:-2] iv) spam[: :-1]	4	L3	3	1
Unit – III					
7. a)	Explain finally keyword with programming example.	4	L2	5	1
b)	What is Matplotlib? Write a Python program to create a bar graph using matplotlib to represent the sales of different products in a store by displaying appropriate labels for the axes and title for the chart. Use different colors for each bar.				

Product	Sales (in units)
Laptop	160
Phone	350
Tablet	200
Headphones	250
Smartwatch	140

8 L3 5 2

- c) Write a Python program that performs division of two numbers. The program should handle ZeroDivisionError and ValueError (if the input is not a number) and print custom error messages. 4 L3 5 2
8. a) What is Pandas in Python? Write a Python program to create a Pandas DataFrame from the following dictionary and access the third row of the DataFrame.
 data={'Name': ['Alice', 'Bob', 'Charlie', 'David'], 'Age': [25, 30, 35, 40], 'Salary': [50000, 60000, 70000, 80000]} 4 L3 5 2
- b) What are Exceptions? Explain the use of try, except, else blocks in Python. Provide examples to show how each block works. 8 L2 5 1
- c) Compare Syntax Error and Exception. Write a Python program that executes an operation on a list and handles an IndexError exception if the index is out of range. 4 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
